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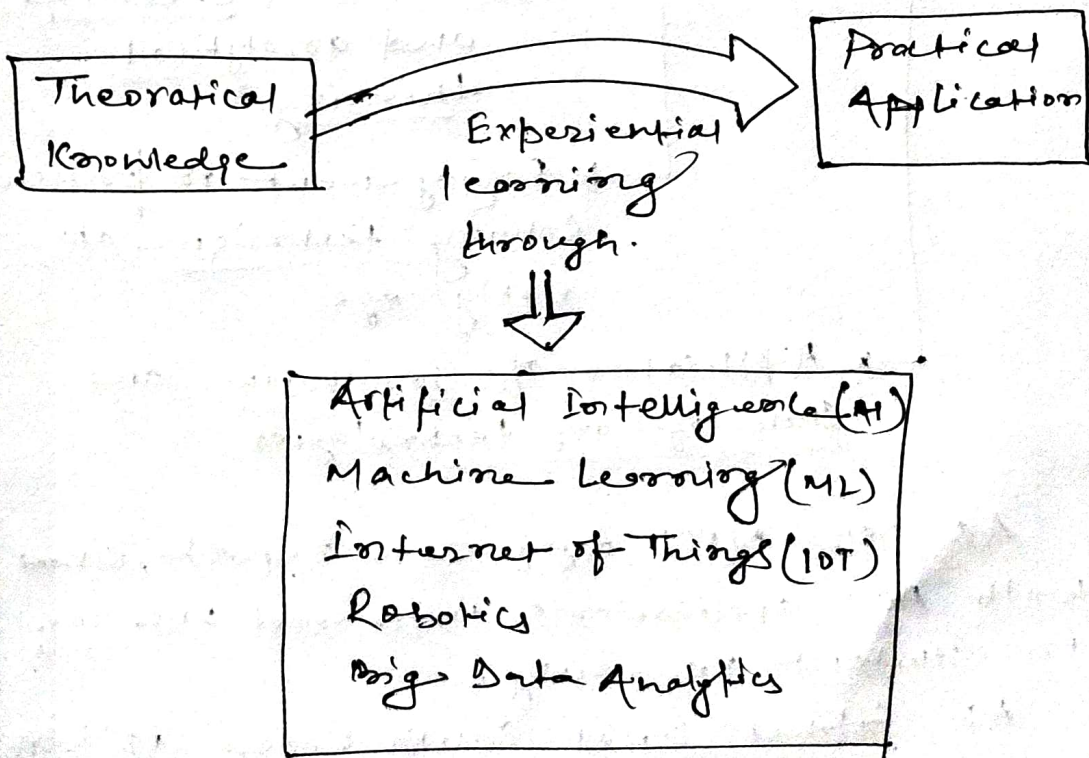
Q. 1.

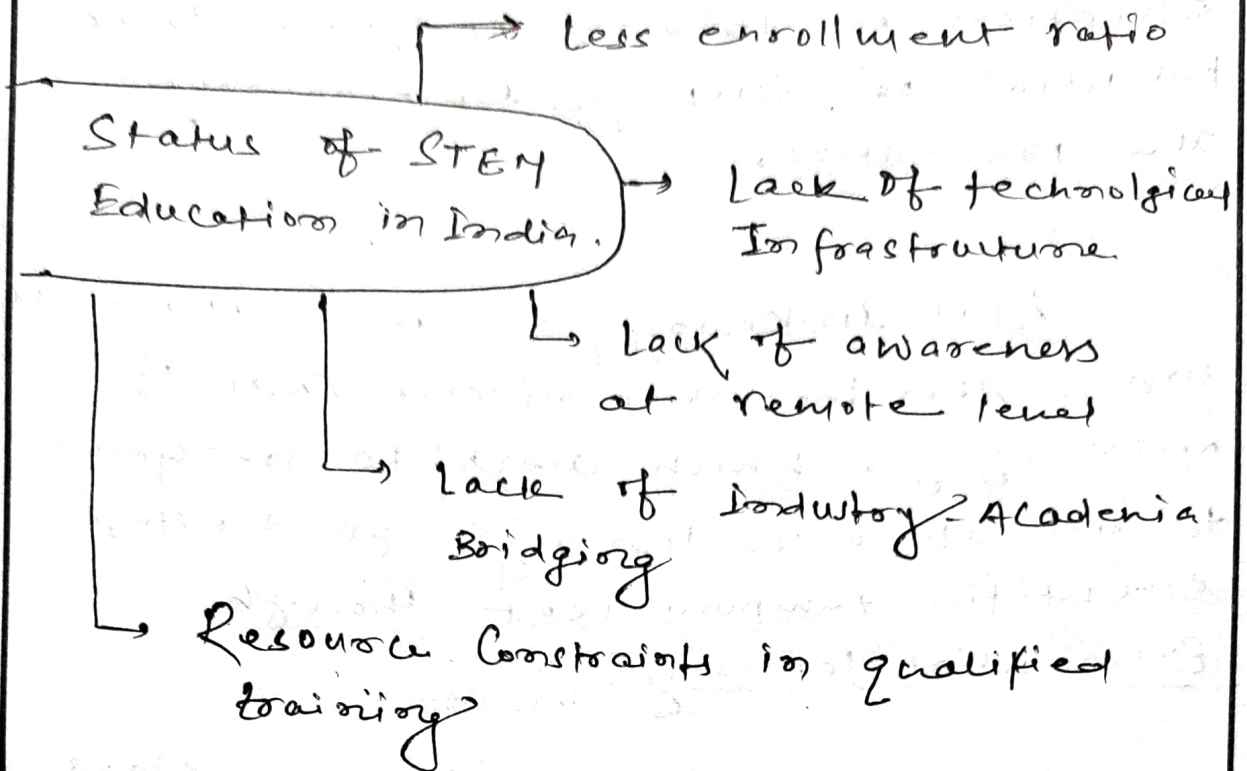
Can Atal Tinkering Labs transform STEM education in India? Analyse their potential on scientific temperament and innovation.

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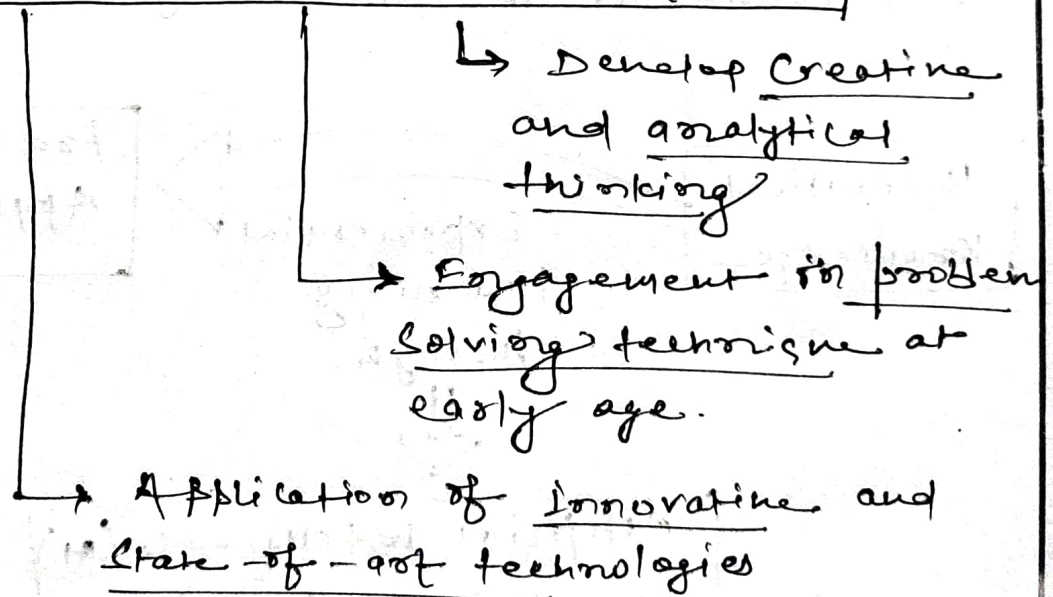
Atal Tinkering Lab is a initiative under Atal Innovation Mission (AIM) of NITI Aayog, which aims to transform traditional rote learning to develop scientific temperament through Experiential learning.

Aim of 50000 Atal Tinkering Labs to be established on Government Schools across India, as a pilot project.





Scope and potential of ATLs:-



As per NITI Aag-2022 report, school with ATL applications improved up to 30% in student reasoning.

At Global level South Korea is best example which transforming their learning to experiential learning at school level.

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Hence, in India, ALE will brought
revolution in school learning through paradigm
shift in education provided efficient
implementation of policy to grassroots
level.

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How Can National Centres of Excellence for Skilling bridge India's skill gap?

Critically analyze the structural and technological challenges they face.

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The initiative of National Centres of Excellence for Skilling is poised for implementing Industry-linked Skilling programs for Indian workforce.

functioning of NCEs.

→ Training India's workforce in emerging and high tech industry as per Global Industry Standards.

→ Industry - Academia Periodic Collaboration.

→ Promotion of R&D in Automation and Robotics.

→ Strengthening Digital Infrastructure.

→ Resource allocation in Skilling.

With these provisions, National Centres of Excellence for Skilling will bridge the India Employment Skill gap in a modern changing Industrial Landscape.

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Structural Challenges

- Traditional rote learning academia
- Resource Constraints
- Infrastructural Support
- Lacking Autonomy in Education Sector
- Industry-Academia Gap

Technological Challenges

- Lack of Res in ~~tech~~ advanced technology sectors.
- Rural-Urban digital divide
- Lack of training in modern technological backdrop.
- Lacking in adoption of Global Industry Standard practices.

Way forward

- Awareness Spreading in experiential learning.
- Prioritizing resource allocation.
- Strengthening infrastructure based on P-P-P model.
- Collab of Industry - Academia,
- Train the trainers program initiation
- focus of R&D in deep-tech.
- Adoption of Global Skill framework.

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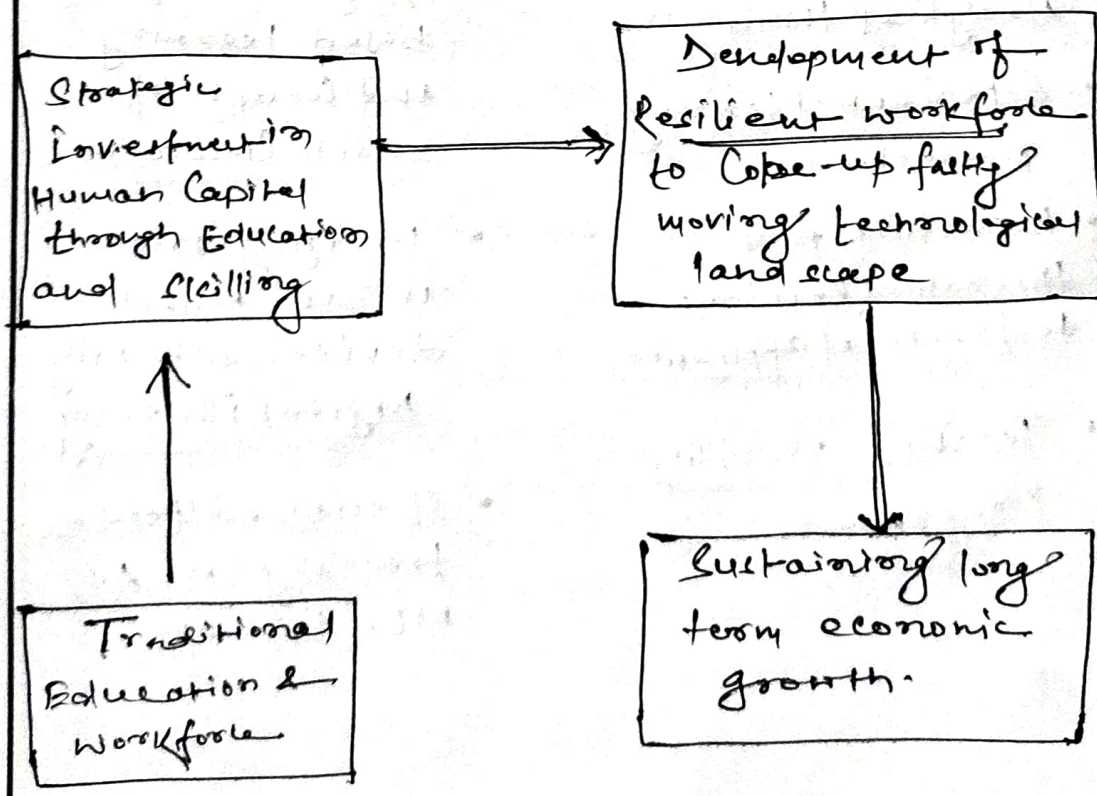
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Discuss the role of education and skill development in Sustainable India's long term economic growth. How can strategic investments in human capital bridge the gap between economic aspirations and realities?

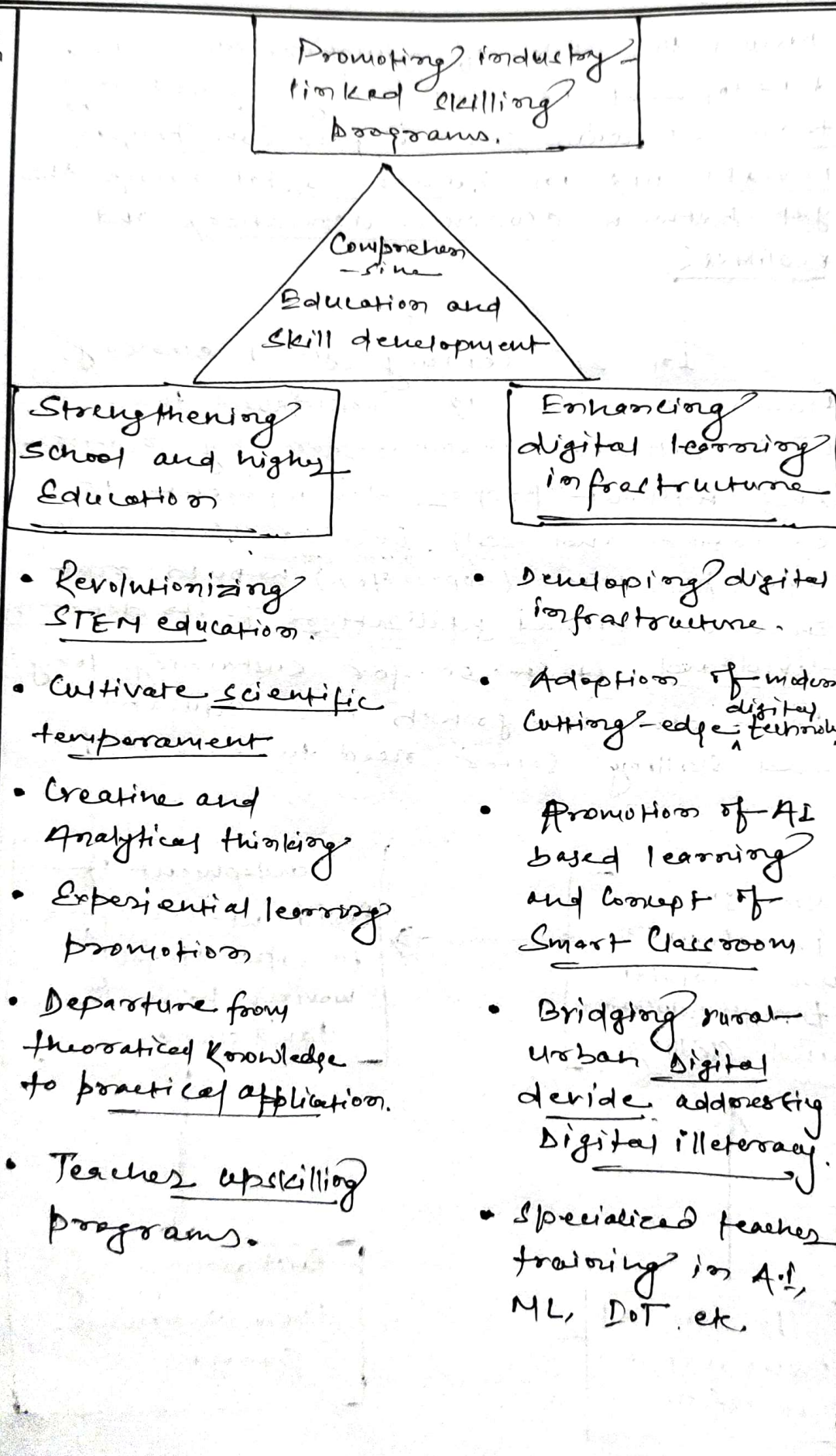
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In 21st century global economy, Human resource is considered the most valuable resource for any nation. And without proper development in education and skill, India's vast working age population (approx 70%) would not ensure optimal utilization of its demographic dividend. It is so for sustainable long term economic growth our education and skilling sector need to be revamped.



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Promoting Industry linked Skill Training Program

- Addressing Skill Gap as per modern industry needs.
- Collaboration of Industry - Academia.
- Vocational and Summer trainings.
- Incentivising theme based Entrepreneurship and Specialized Apprenticeship.
- Adopting Global Standards as per ~~new~~ changing technological landscape.

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Strategic investment in Human Capital

In ~~the~~ bridging the gap between Economic aspiration to realities a Strategic investment need to be done in Human Capital. For this 3 three sectors may be selected & for nurturing Human Capital

Schooling & Higher Education.

- For strengthening this sector Government has initiated establishment of 50,000 Atal Tinkering Labs.
 ↳ It will foster Creative & analytical thinking development at early age.

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- Bhasha Bhasa Pustak Scheme

↳ will develop digital books in Indian languages.

↳ Education in mother language would foster Cognitive development

As per World Language Report-2023 :

Students taking early schooling in their mother language have shown 40% higher learning ability.

- Expanding IIT Infrastructure

↳ will promote R&D.

↳ Foster research in Deep-tech and Cutting edge tech domains

↳ India expends only 0.67% of GDP in R&D as behind China (2.4), US (3.1) & South Korea (4.5%).

Enhancing Digital Learning Infrastructure

- Incentivising and promoting Edtech Startup Ecosystem.

- Developing Smart Classroom through micro-investment at remote locations

- Developing of Centre of Excellence in AI for integrating AI in education

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- Bridging ~~Old~~ Digital Divide through P-P-P in infrastructure development.

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Bridging Skillings Gap

- ↳ Incentivising Sector Specific Apprenticeship program.
- ↳ National Centres for Excellence for Skillings
 - Implementing Industry-linked Skillings program
 - Training India's workforce in emerging and high tech industry.
- ↳ 4% per NEF: The future Jobs 2023
India would produce 85 million new jobs in digital & automation if strategically investment done.

At last, assessing the vital role of education and skillings, a strategic investment in human capital may prove game changes for Indian economy. But the transformative potential of this will depend mostly ^{on} effectiveness of state machinery in implementing these policies and schemes.